When Realized Outcomes Outweigh Predictive Signals: Evidence from Fantasy Premier League

Abstract

Large crowds can aggregate dispersed information while responding disproportionately to new signals. Fantasy Premier League makes both properties observable because the platform publishes the share of entries holding each player. Using 100,801 player-gameweeks across four seasons, we show that aggregate ownership predicts subsequent performance beyond predetermined public-information benchmarks. Yet ownership changes reject a proportional-calibration benchmark: realized scoring receives a much larger demand response than signals with greater predictive content. Among players matched on position, price, prior ownership, playing time, and match-level non-penalty expected goals, scorers are followed by 1.3 percentage points more ownership without detectable differences in subsequent chance quality or points. Adding ownership to the benchmark does not detectably improve out-of-sample static squad selection. Aggregate wisdom and marginal miscalibration can coexist.

Keywords: fantasy sports; information aggregation; belief updating; consumer demand; behavioral economics

JEL Classification: D83, D84, D91, L83, Z20

1. Introduction

Large crowds often aggregate dispersed information effectively. But informative aggregate positions need not imply proportional responses to new signals. A population can hold useful positions while responding disproportionately to the next realization. Separating these properties requires observing both the stock of collective choices and the flow of subsequent reallocation, two objects that standard market prices and most experiments do not jointly reveal.

Standard market data primarily reveal prices, which combine beliefs with liquidity, risk appetite, and supply. Experiments can elicit beliefs directly, but rarely observe repeated allocation by a naturally occurring population at this scale.

Fantasy Premier League recorded more than 11.5 million registered entries in 2024-25 and publishes the percentage of entries holding every player. Participants face a common roster structure and begin under the same budget constraint. Collective ownership is therefore observed directly as a quantity rather than inferred from a market price. The platform also records a compact set of dated public performance signals whose predictive content can be estimated on the same panel that records aggregate reallocation.

Aggregate ownership contains incremental information. Using the previous-gameweek archival snapshot, which necessarily predates the current gameweek's outcome, the ownership-rank coefficient is 0.831 after conditioning on a public-information benchmark and player and gameweek fixed effects. The ownership coefficient remains positive across alternative benchmarks and appears again in an untouched validation season.

The crowd nonetheless does not reallocate in proportion to measured predictive content. We compare the subsequent demand response to each public signal with that signal's own predictive content, using expectations formed only from prior information. The proportional-calibration benchmark is strongly rejected across all ten signals and across the eight signals that enter fantasy scoring. Playing time carries more predictive content than any other signal we examine but moves ownership by far less than a goal does; goals carry less predictive content than playing time or either chance measure but generate the largest demand response.

The mismatch is clearest around realized scoring, and documenting it does not require the points-forecasting benchmark. Among players matched on position, price, prior ownership, playing time, and match-level non-penalty expected goals, scoring is followed by a 1.30 percentage-point larger ownership increase. The matched groups show no detectable difference in subsequent chance quality or points, while timestamped transfers place the response primarily on the acquisition side.

These results do not identify a psychological mechanism or yield a detectable gain in our static selection exercise. Adding ownership to the benchmark does not detectably improve out-of-sample squad selection, and attention remains an interpretation rather than a finding.

The paper makes two contributions. First, it separates the information contained in aggregate holdings from the calibration of marginal updating, two properties usually observed through the same market price. Second, it provides field evidence that realized sports outcomes can receive disproportionate demand relative to signals with greater predictive content. Fantasy sports make the two objects jointly observable because the platform records repeated quantity choices at high frequency under common allocation constraints.

2. Related Literature

2.1 Fantasy sports, collective judgment and observed quantities

Markets typically reveal prices rather than quantities, making it difficult to separate beliefs from liquidity, risk appetite, and supply (Wolfers & Zitzewitz, 2004). This setting sidesteps that problem because ownership is a quantity published directly rather than inferred from a price.

The closest design precedent is Losak et al. (2023), who find no hot-hand predictability in daily fantasy baseball even though participants pay for recent performance, and no simple rule that profitably exploits the gap; the question follows Gilovich et al. (1985). Coates and Parshakov (2022) show that crowd-sourced footballer valuations are correlated with, but systematically below, realized transfer fees. O'Brien et al. (2021) document persistent managerial skill within Fantasy Premier League. Our weekly ownership panel adds a different object: repeated aggregate quantity choice. That distinction lets us separate what the crowd already holds from how those holdings change when new information arrives.

2.2 Outcome realization and underlying performance

Goals are rare enough, and the gap between performance and scoreline wide enough, that Brechot and Flepp (2020) show expected-goals measures rank teams more informatively than results do. Separating an outcome from the process producing it is therefore the central measurement problem in football. Deviations from expected performance contain both components: Holzmeister and Johannesson (2025) decompose them across European leagues and attribute roughly two fifths to skill. Conditional on the chances a player receives, whether the ball goes in reflects both luck and skill. Our design studies the crowd's response to that realization without assigning the residual to either component.

Gauriot and Page (2019) provide the closest identified evidence, comparing narrowly separated scoring and non-scoring attempts around the goal frame and finding that evaluators reward the realized outcome beyond the performance behind it; Kausel et al. (2019) provide related evidence from subjective performance ratings. Appendix M.2 implements a narrower goal-versus-woodwork comparison where shot-event data are available, while the broader matched comparison remains our main behavioral evidence. Related work separates outcomes or popularity from underlying performance in sports prices: Flepp et al. (2024) show that betting exchanges overvalue teams that previously overperformed their expected goals, and Jedelhauser et al. (2023) show that footballer valuations can reflect popularity beyond marginal contribution. What this setting adds is the object being reallocated: not an evaluator's rating, but the aggregate quantity a population chooses to hold, observed alongside the opportunities that player subsequently receives.

2.3 Information weighting and behavioral sports markets

Behavioral theories give several reasons why response weights may depart from predictive content. Salient events can receive disproportionate weight (Bordalo et al., 2012), and limited attention governs which public information is acted upon (Hirshleifer & Teoh, 2003). Augenblick et al. (2025) provide a closely related implication: decision makers can over-infer from weak signals and under-infer from strong ones. Testing relative information weighting requires observing both the signals available to decision makers and an empirical benchmark for their subsequent predictive content, and Fantasy Premier League supplies both.

The buy-sell decomposition connects to evidence that attention can operate asymmetrically across acquisition and disposal. Barber and Odean (2008) show that individual investors are net buyers of attention-grabbing assets, because buying requires search across a large choice set while selling is confined to assets already held. Recent sports-market work documents related behavioral distortions in football ticket demand (Kowalik & Pawlowski, 2026), while Winkelmann et al. (2024) show that apparent sports-market inefficiencies can arise by chance in short samples. The latter motivates our use of calibrated nulls and an untouched validation season.

Prior work therefore establishes each side of our question separately: crowds can contain useful information, and decision makers can overweight realized performance. We ask whether both properties coexist within the same large repeated-choice population, and which signals generate the divergence.

3. Institutional Setting and Data

3.1 Rules and timing

Fantasy Premier League is the official free-to-enter fantasy competition of the English Premier League. Participants assemble squads of real footballers and earn points from their match performance. Each entry begins under the same budget and roster rules, although subsequent team value can evolve as administered prices change. The competition runs across 38 gameweeks from August to May, where a gameweek is the block of fixtures between successive squad deadlines. Table 1 summarizes the rules relevant to our estimation period.

Prices are administered rather than market clearing. A player's price is set by the platform and can move by 0.1m in response to transfer activity under an undisclosed formula. Because every entry can hold the same player, changes in ownership are observed separately from changes in the administered price.

Ownership is published rather than inferred. The reported percentage is the share of all entries registered for the season, active or not, so a squad whose owner has stopped managing still counts toward it. It measures whether a player is held, not whether he is captained; captaincy increases the value of holding high-scoring players but does not change the ownership measure itself. Participants can also use single-use chips that temporarily alter transfer or scoring constraints. These affect the number or structure of decisions but not the definition of ownership used here.

Entry is free, so participants differ in effort and skill, but they face the same roster structure and begin from the same budget rule. Aggregate ownership is therefore a headcount over a common constrained choice problem rather than a wealth-weighted market position. Ownership is displayed continuously, whereas squad decisions lock before each gameweek. Because the historical archive does not identify the exact capture moment within the pre-deadline window, Section 5.1 leads with a previous-gameweek snapshot that necessarily predates the current outcome; Appendix G diagrams the timing.

[TABLE 1 ABOUT HERE]

3.2 The estimation panel

The estimation panel contains 100,801 player-gameweeks from 2021-22 through 2024-25, after aggregating double-gameweek fixtures to a player-gameweek key and applying the two exclusions documented in Appendix G. Table 2 reports summary statistics.

Two objects must be distinguished. Aggregate ownership is the stock: the share of registered entries holding a player. The demand response is the flow: the change in that share from one gameweek to the next, together with the individual transfers that produce it. Because the number of registered entries changes within a season, a change in ownership share is not identical to net transfers; Section 3.3 uses the individual panel to decompose actual acquisitions and disposals. The information analysis uses the stock, while the calibration and matched analyses use the flow.

Ownership is heavily right skewed, so it enters the information regressions as a within-gameweek percentile rank, from 0 for the least held player to 1 for the most held. The behavioral specifications instead use the change in the ownership share, measured in percentage points of registered entries.

[TABLE 2 ABOUT HERE]

3.3 Individual transfer panel

Individual behavior is observed for one season on 999 participants, comprising 91,364 transfers timestamped to the second and 509,610 manager-player-gameweek holding records. The subsample was drawn from the upper end of the overall standings rather than at random, and is accordingly unrepresentative: median final standing is 864 and the quartiles are 283 and 2,894. The panel is therefore used only to corroborate timing and the buy-sell margin, not to estimate population-wide prevalence.

3.4 Signals and outcomes

The calibration analysis uses ten post-match signals observed during the estimation period: goals, assists, bonus points, clean sheets, saves, yellow cards, playing time, the bonus point index, and the platform's expected goals and expected assists. The eligibility rule is that a signal be a participant-facing platform quantity available throughout the estimation period. Eight of the ten directly award, deduct, or determine fantasy points; expected goals and expected assists are published by the platform but do not enter any participant's score. Section 5.2 also reports a restriction to the six signals that award or deduct points at every position: goals, assists, bonus points, the bonus point index, playing time, and yellow cards. Saves and clean sheets are excluded because they do not: only a goalkeeper records saves, and a clean sheet pays four points to a goalkeeper or defender and one to a midfielder but nothing to a forward. The criterion is whether the signal pays out at every position, not whether it pays the same amount at each, which for goals it does not. The platform began publishing its expected-goals measures in 2022-23, so the calibration sample covers three of the four estimation seasons and every signal is estimated on that common window, standardized within gameweek. Appendix B states both restrictions and Appendix L documents availability, first-stage performance and the quantities excluded as exact combinations of the others.

Two expected-goals series appear in the paper and should not be conflated. The platform's own expected-goals measures enter the calibration analysis and include penalties. The matched design of Section 4.4 instead uses match-level non-penalty expected goals from Understat, because the object of interest there is realized attacking opportunity rather than fantasy scoring itself. That series covers 72.7 percent of appearances; Appendix G documents the join and its validation checks.

Outcomes are the realized points award, and the sums of points and of non-penalty expected goals over the following one, three and five gameweeks, constructed within a player and season. A horizon of three gameweeks means gameweeks $t + 1$ through $t + 3$ and never includes the event gameweek t.

4. Empirical Strategy

The empirical strategy follows the paper's four questions in sequence. Sections 4.1 and 4.2 construct the public-information benchmark and ask whether aggregate ownership contains information beyond it. Section 4.3 compares the predictive content of public signals with the demand responses they generate. Sections 4.4 and 4.5 focus on realized scoring without relying on the points forecast. Section 4.6 then asks whether the information contained in ownership improves a static legal-squad selection rule.

4.1 The public-information benchmark

The benchmark is an expectation of a player's gameweek points formed only from information available beforehand. It uses 36 predetermined variables covering recent playing time, starting and benching history, rolling expected goals and assists, the bonus index, opponent strength, implied team goals, home advantage, fixture congestion, price, and position. Ownership and transfer variables are excluded, as is any feature constructed from a full player-season aggregate. We estimate the benchmark using gradient-boosted trees (Friedman, 2001) in a walk-forward design: each gameweek is predicted using only earlier observations, and hyperparameters are fixed independently of the study outcomes. Appendix H lists the variables and the information-set checks.

Section 5.1 also considers four simpler benchmarks: bookmaker match odds, recent form, the administered price, and a benchmark stripped of price, ownership, and transfers, which we call FPL-demand-free. The last still contains bookmaker odds and is therefore not free of all crowd-derived information. Price is reported separately because it responds to transfer activity by rule.

4.2 Information contained in aggregate ownership

For the first question we estimate

$$y_{it} = \beta\, r_{it} + \theta\, q_{it} + \alpha_i + \delta_t + \varepsilon_{it}, \tag{1}$$

where y_{it} is the points player i is awarded in gameweek t, or the sum over gameweeks $t+1$ through $t+3$ or $t+5$; r_{it} is the within-gameweek rank of aggregate ownership; q_{it} is the within-gameweek rank of the public-information benchmark; and α_i and δ_t are player and gameweek fixed effects. Both ranks run from 0 for the least held or lowest forecast player in a gameweek to 1 for the highest, and θ is a nuisance coefficient on the benchmark rank. With both effect sets absorbed, β is identified from variation in a player's ownership rank around his own average within a gameweek, and it is the gap in points between the least and most held player. Including both ranks makes β

the incremental predictive association of ownership conditional on the benchmark, rather than the predictive association of ownership taken in isolation. The symbol γ_j is reserved throughout for the predictive content of a signal innovation in Equation (3).

4.3 Signal innovations and the proportional-calibration benchmark

A signal's level is not news. The relevant object is its innovation: the realized signal less what could have been expected beforehand. We form that expectation by forward chaining, fitting the model for each gameweek on earlier gameweeks only and predicting the current gameweek from predetermined variables. Random-fold cross-fitting is unsuitable here because it would allow later observations to inform an earlier-period expectation. We retain only signals for which this forward-chained first stage outperforms a running-mean predictor out of sample. The criterion uses first-stage performance only and does not inspect subsequent ownership or future points.

For each signal innovation we estimate two coefficients on the same sample and with the same controls, all innovations entering jointly in both equations: its association with subsequent ownership adjustment, and its incremental predictive content for points over the following three gameweeks. Let u_{it}^j denote the standardized innovation in signal j. We estimate

$$\widetilde{\Delta o}_{it} = \sum_j \beta_j u_{it}^j + X_{it}'\eta + \varepsilon_{it}, \tag{2}$$

$$\widetilde{Y}_{it}^{(3)} = \sum_j \gamma_j u_{it}^j + X_{it}'\rho + \nu_{it}, \tag{3}$$

where $\widetilde{\Delta o}_{it}$ is the standardized change in the ownership share from t to $t+1$, and $\widetilde{Y}_{it}^{(3)}$ is standardized points over gameweeks $t+1$ through $t+3$. Each coefficient is therefore the standard-deviation change in its own outcome associated with a one-standard-deviation signal innovation, which is what makes β_j and γ_j directly comparable. The underlying ownership variable is the change in the share, not a change in rank.

The control vector X_{it} contains a constant, the within-gameweek price rank, and the within-gameweek existing ownership rank. These absorb two dimensions of a player's existing market position before the new signal is incorporated. Because innovations are standardized within gameweek, the baseline specification does not include separate gameweek effects; Appendix L.4 reports the control ladder and shows that adding player and gameweek fixed effects leaves the response coefficients nearly unchanged.

Proportional calibration implies a simple restriction. If participants respond to each signal with a common linear responsiveness to the revision in expected points it implies, the demand response is proportional to the predictive content with the same constant for every signal,

$$\beta_j = \lambda \gamma_j \quad \text{for every } j, \tag{4}$$

Under this restriction, all signal-response pairs lie on a common ray through the origin and λ is the only free parameter. The restriction is a diagnostic benchmark for whether aggregate reallocations scale with measured predictive content; it is not a structural characterization of optimal play under the full set of Fantasy Premier League constraints, and not the unique definition of rational choice. We therefore interpret rejection as disproportionate response relative to measured three-gameweek predictive content. The omnibus test establishes whether responses depart from proportional calibration; the residuals then identify which signals receive relatively more or less demand response. Appendix B states the benchmark and reports the omnibus test.

4.4 Matched realization comparison

To remove the points benchmark from the central comparison altogether, we group player-gameweeks within a season on five exact strata: position; price rounded to the nearest half million; a within-gameweek tercile of prior ownership; a playing-time band, either 45 to 80 minutes or above 80; and a within-gameweek quantile bin of match-level non-penalty expected goals. Only players appearing at least 45 minutes are eligible, and only groups containing both a player who scored and a player who did not are retained.

We choose 40 expected-goals bins using balance alone, before examining the ownership or future-performance contrasts. The standardized difference in non-penalty expected goals falls from 0.539 with five bins to 0.090 with forty; moving to eighty bins improves balance only to 0.080 while reducing the retained sample by roughly one third. Because binning does not fully absorb a variable this skewed, the outcome regressions also include non-penalty expected goals linearly and quadratically. The contrast is a difference of group means, with group and gameweek effects absorbed and standard errors clustered on player and on gameweek.

The matching variables are realized within the event match, not predetermined treatment covariates. Scoring can alter game state, subsequent opportunities and substitution timing. The design therefore compares players with similar measured realized opportunity; it does not create random assignment of scoring. Appendix E reports the diagnostics and states the limits.

4.5 The individual transfer design

The individual panel allows us to locate the adjustment in time and to separate buying from selling. The unit is a player-gameweek, restricted to appearances of at least 45 minutes. Players are grouped within a team-fixture, meaning one side of one match, together with bands for price, prior ownership, and playing time, retaining only groups containing both a scorer and a non-scorer. Players compared within a group were therefore teammates in the same match facing the same opponent, so the attacking environment is held fixed by construction. The design yields 415 groups covering 1,438 player-gameweeks.

Transfers are attributed to the most recent preceding match and decomposed by time measured from kickoff rather than from the end of play, with a seven-day cap preventing double attribution. Counts enter in logarithms. Buy and sell rates are formed against the population able to take each action, since only a holder can sell while everyone else can buy: the buy rate divides acquisitions by the share not holding the player and the sell rate divides disposals by the share holding him. Group effects are absorbed and standard errors are clustered on player. Appendix N gives the windows.

Standard errors elsewhere are clustered two ways, on player and on gameweek, following Cameron et al. (2011). A wild-cluster bootstrap for the headline Equation (1) coefficient yields $p = .001$, and overlapping horizons and the calibration statistic use a gameweek block bootstrap. Appendix J reports the details. The study was not preregistered; specifications were frozen after exploratory diagnostics and before the reported robustness and validation analyses were run.

4.6 Static legal-squad selection

The final question is whether the incremental information in ownership is large enough to change a feasible squad. For each gameweek of the evaluation season, we solve by integer programming for a fifteen-player squad and legal starting eleven subject to the platform's budget, club, position, and formation constraints. We compare three scoring rules: the public-information benchmark alone, ownership rank alone, and the benchmark plus a weight on ownership rank. The ownership weight is chosen on the first three seasons over a grid that includes negative values and then applied unchanged to the evaluation season. Performance is realized points per starting slot, and the arms are compared gameweek by gameweek.

This is a static exercise. It ignores the existing squad, transfer continuity and penalties, captaincy, the bench and price changes, all of which the dynamic problem contains. It is not a playable season-long strategy.

5. Results

5.1 Aggregate ownership contains incremental information

Aggregate ownership predicts subsequent performance beyond the public-information benchmark. Because the archive records a gameweek-boundary snapshot rather than a capture timestamped at a known moment, our conservative specification uses the previous-gameweek archival snapshot, which was taken before the current match was played; that timing rules out contamination by the current gameweek's outcome. In that specification the coefficient on ownership rank is 0.831 (t = 11.8) with player and gameweek fixed effects. Because the rank runs from zero to one, the coefficient is the predicted points difference between the least and most held player within a gameweek, conditional on the benchmark, against a sample mean award of 1.205 (Table 2). Same-gameweek archival ownership produces a larger estimate of 1.403. What the crowd already holds carries information about what has not yet happened.

Adding ownership rank to the benchmark's feature set raises the out-of-sample R^2 of a forward-chained forecast of the points award from 0.3210 to 0.3221. The improvement in overall predictive fit is small in absolute terms, consistent with ownership containing detectable but limited information beyond an already rich public-information benchmark. The ownership coefficient is stable across alternative benchmarks and lies outside every draw of two permutation nulls (Appendix J). The benchmark itself excludes ownership but contains the administered price, which responds to transfer activity by rule, so it is ownership-free without being demand-free. Against the FPL-demand-free benchmark, which excludes ownership, transfers, and price, the coefficient is 1.496.

The positive association also appears in the untouched 2025-26 validation season. Conditional on price rank, the ownership coefficient is 0.954 (t = 2.5) on 29,137 player-gameweeks. Because the platform revised its scoring rules and the full benchmark cannot be reconstructed for that season, the magnitude is not directly comparable with the estimation sample. We therefore treat this as external validation of the positive ownership-performance association rather than as a replication of the full design.

[TABLE 3 ABOUT HERE]

5.2 Demand responses reject proportional calibration

The proportional-calibration benchmark is strongly rejected: $W = 111.0$ on nine degrees of freedom, $p < .001$, with a best-fitting multiple of 0.226 (standard error 0.073).

The rejection is not driven by the two signals that do not enter fantasy scoring. Restricted to the eight point-relevant signals the statistic is 97.6 on seven degrees of freedom, and restricted further to the six signals that award or deduct points at every position it is 84.3 on five, both with $p < .001$. We test proportionality jointly rather than through signal-by-signal ratios, because several predictive-content coefficients approach zero and make those ratios unstable. The rejection survives removing any one signal and leaving out any one season. No individual comparison with goals survives the Benjamini and Hochberg multiplicity adjustment, so formal inference rests on the omnibus proportionality test rather than on any single pairwise contrast (Appendix J).

The economic magnitude of the rejection is easiest to see in the signal pairs. A one-standard-deviation playing-time innovation predicts 0.129 standard deviations more points over the following three gameweeks but is associated with only 0.024 standard deviations more ownership change. The corresponding predictive-content coefficients are 0.058 for chance quality and 0.064 for chance creation, with ownership responses near zero. Goals, by contrast, carry only 0.030 standard deviations of predictive content but generate a 0.112-standard-deviation ownership response. Among these four signals, goals have the smallest predictive coefficient and the largest demand response. Table 4 reports the pair for every signal.

Figure 1 displays the full set against the fitted proportional ray. Across the ten innovations the rank correlation between response and predictive content is -0.418, with a 95 percent interval from -0.515 to 0.042. We therefore do not establish a negative ranking, but we find no reliable positive ordering of demand response by predictive content. The signals that move the crowd most are not the signals that predict most.

[TABLE 4 ABOUT HERE]

[FIGURE 1 ABOUT HERE]

5.3 Scoring is followed by higher ownership without detectable improvement in subsequent chance quality

Within the matched opportunity set, players who score are followed by a 1.297 percentage-point larger increase in ownership (t = 8.5). Scoring is not randomly assigned, so this is matching, not random assignment, and we make no causal claim. The matching nevertheless achieves close balance on measured opportunity: the standardized difference in match-level non-penalty expected goals is 0.090, and no other matching covariate exceeds 0.10. Two player-gameweeks that looked alike in measured chance quality before the match are therefore followed by different ownership according to which one converted.

Subsequent chance quality is not detectably different. Over the following three gameweeks the contrast is -0.003 standard deviations, with a 95 percent interval from -0.059 to 0.054, so any improvement above 0.054 standard deviations can be excluded. The corresponding point differences over the following one, three, and five gameweeks are 0.14, 0.06, and 0.25 fantasy points, none detectably different from zero. These results bound subsequent measured opportunity; they do not rule out persistent differences in finishing, role, or other dimensions that non-penalty expected goals does not capture.

Balance at this partition costs sample: the matched groups retain 4,897 of 21,573 eligible player-gameweeks, 22.7 percent, including 72.0 percent of eligible scoring observations. Coarser partitions retain more at worse balance and give similar or slightly larger contrasts (Appendix K). Table 5 reports three alternative constructions.

[FIGURE 2 ABOUT HERE]

Figure 2 traces both quantities week by week. Event time is indexed by τ, with $\tau = 0$ the gameweek containing the match and $\tau = -1$ the omitted period against which ownership is measured. We do not detect pre-event divergence (W = 1.83 on two degrees of freedom, bootstrap p = .41). The ownership contrast appears in the first post-event gameweek and widens thereafter, while measured chance quality remains approximately flat, never exceeding 0.056 standard deviations in absolute value. We use this pattern as a diagnostic of pre-event comparability. It is not difference-in-differences identification.

[TABLE 5 ABOUT HERE]

The contrast is present in every season taken on its own, running from 0.849 to 1.547, and survives leaving any one season out. Replacing expected goals with raw shot and key-pass counts, so that no expected-goals model enters the grouping, leaves the demand contrast large, although its subsequent-opportunity placebo does not hold in that specification (Appendix K).

A narrower comparison sets converted attempts against attempts that struck the woodwork. It yields a positive ownership response and no detectable difference in subsequent chance quality, but scorers in that comparison earn more subsequent points (Appendix M.2). Scoring therefore carries information about subsequent scoring that measured chance quality does not. We treat the comparison as corroboration of realization-sensitive reallocation rather than as causal evidence of selectively miscalibrated updating, and we do not interpret the matched coefficient as the causal effect of the ball crossing the goal line.

5.4 The response arrives quickly and through buying

The individual panel locates the adjustment primarily on the acquisition side. Within team-fixture matched groups, acquisitions of scorers rise within six hours of kickoff, where the log contrast is 0.246 (t = 6.7). A goal innovation is associated with a 0.090 log-point increase in the normalized acquisition rate (t = 4.9), while the normalized disposal response is 0.005 (t = 0.4); both rates enter in logarithms. Because this panel contains 999 highly ranked participants rather than a representative sample, we use it only to corroborate the timing and buy-sell margin of the aggregate response. Appendix N reports the full timing windows.

5.5 Adding ownership does not detectably improve out-of-sample squad selection

In the evaluation season the public-information benchmark produces 4.823 points per starting slot, against 4.484 for ownership alone. The paired difference is -0.339 across 37 gameweeks ($t = -1.70$). The benchmark therefore has the higher sample mean, but the difference is not estimated away from zero and does not establish that ownership is the inferior selection score.

The relevant comparison is whether ownership adds value conditional on the benchmark. The combined rule produces 4.855 points per starting slot, only 0.032 above the benchmark alone ($t = 0.21$). We therefore detect no improvement from adding ownership. The bound is not tight: at 80 percent power for a two-sided 5 percent test the minimum detectable effect is approximately 0.43 points per starting slot, so gains below that threshold cannot be ruled out. The tuning grid included negative ownership weights, but no contrarian tilt was selected; the training seasons selected a weight of 0.4. The information in aggregate ownership is therefore real but not, on this margin, worth acting on. Appendix M.4 reports the full legal-squad exercise and weight profile.

6. Interpretation

6.1 What the evidence establishes, and what it does not

The paper's two central results can appear contradictory. Aggregate ownership is informative across the benchmark specifications we consider, yet marginal demand responses are not proportional to measured predictive content. Being informative in levels and miscalibrated in updating are different properties of the same population, and there is no reason they must move together.

The distinction is between a stock and a flow. Aggregate ownership is a stock accumulated from many prior signals; weekly reallocation is a flow responding to new information. The stock can therefore contain substantial accumulated information even when the individual updates entering it are not proportionally calibrated, because the level is dominated by what the crowd already knew. Nor does miscalibration imply a selection rule exploitable in the static legal-squad exercise: adding ownership to the benchmark does not detectably improve it, and directly discounting the realization-sensitive component performs no better out of sample (Appendix M.5). We therefore describe the finding as selectively miscalibrated updating by an informative crowd, not as irrationality.

6.2 Mechanism and limits

One candidate explanation is salience: goals are conspicuous realized outcomes in a way that playing time and chance quality are not, and the acquisition-heavy response is consistent with it. But the mechanism is not identified. Recency weighting could generate a similar pattern without a distinct attention channel, and a scheduling-based proxy for audience attention moves in the opposite direction from the predicted mechanism (Appendix M). The near-goal comparison does not isolate miscalibrated updating either: scorers there earn more subsequent points despite no detectable difference in future non-penalty expected goals, so scoring may reveal predictive information that measured chance quality does not capture. We do not interpret the matched coefficient as the causal effect of the ball crossing the goal line, and we leave the mechanism open.

Four limitations bound the interpretation. The matched comparison is observational and applies only to the retained common-support sample, because playing time and opportunity are measured in the event match. The transfer timing and buy-sell decomposition come from highly ranked managers and need not generalize to the full population. Ownership is a constrained allocation rather than an elicited belief. And administered prices rise with net transfers, so some acquisitions may anticipate a price increase rather than a revision in expected points; the static exercise of Section 5.5 does not model price appreciation and cannot rule that motive out.

7. Conclusion

Aggregate ownership in Fantasy Premier League contains predictive information beyond public benchmarks, but subsequent reallocations are not proportional to the predictive content of new signals. Realized scoring is followed by especially strong demand conditional on measured opportunity quality, while adding ownership to a static squad-selection benchmark yields no detectable improvement. The matched scoring comparison remains observational, and the evidence does not identify an attention mechanism. The central result is therefore narrower: aggregate holdings can contain useful information even when marginal responses to new signals are selectively miscalibrated. An informative stock and a miscalibrated flow can coexist.

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Table 1

Fantasy Premier League Rules and Analysis Samples

Element	Value
Initial squad	100m budget; 15 players (2 GK, 5 DEF, 5 MID, 3 FWD), at most 3 per club
Legal eleven	1 GK, 3 to 5 DEF, 2 to 5 MID, 1 to 3 FWD
Transfers	1 free per gameweek, banked to a cap of 2 through 2023-24 and 5 after; extras cost 4 points
Scoring	Goals GK 6, DEF 6, MID 5, FWD 4; assists 3; clean sheet 4 for GK and DEF, 1 for MID; 1 point to 60 minutes, 2 above; up to 3 bonus points from the Bonus Points System; captain doubles
Deadline	90 minutes before the gameweek's opening fixture
Ownership	Share of all registered entries holding a player, active or not; displayed continuously
Estimation panel	100,801 player-gameweeks, 866 players, 146 gameweeks, 2021-22 to 2024-25
Individual panel	91,364 timestamped transfers, 999 participants, 1 season
Validation season	29,137 player-gameweeks, 840 players, 2025-26, revised scoring, Section 5.1 only

Note. Monetary values are the platform's internal budget units and scoring entries give points awarded. Rules are those of the estimation seasons; 2025-26 revised the scoring and enters only Section 5.1.

Table 2

Summary Statistics, Estimation Sample

Variable	N	Mean	SD	p10	Median	p90
Points	100,801	1.205	2.465	0.000	0.000	4.000
Ownership share, percent	100,801	3.022	8.734	0.013	0.259	7.504
Change in ownership, percentage points	97,637	0.008	1.833	-0.240	-0.001	0.172
Price, millions	100,097	4.928	1.104	4.000	4.500	6.000
Minutes	100,801	28.719	41.119	0.000	0.000	90.000
Non-penalty expected goals	30,306	0.105	0.232	0.000	0.000	0.362
Expected assists	30,306	0.080	0.182	0.000	0.000	0.282
Shots	30,306	0.904	1.319	0.000	0.000	3.000

Note. One observation is a player-gameweek. Ownership is the share of registered entries holding the player; its change to the following gameweek is the outcome of the behavioral specifications. The three opportunity measures are match-level Understat figures over the appearances where that series exists (Section 3.4). Price is missing for 704 rows, which enter no specification using it.

Table 3

Incremental Predictive Content of Aggregate Ownership

Specification	Ownership coefficient	t
Previous-gameweek snapshot, public-information benchmark	**0.831**	**11.8**
Same-gameweek snapshot, public-information benchmark	1.403	20.4
Same-gameweek snapshot, FPL-demand-free benchmark	1.496	20.8
Same-gameweek snapshot, administered-price benchmark	3.010	38.3
2025-26 validation season, administered-price benchmark	0.954	2.5

Note. Each row is the coefficient β on ownership rank in Equation (1), with the gameweek's points award as the outcome. All rows carry player and gameweek fixed effects. Rank runs from 0 for the least held player to 1 for the most held, so the coefficient is the gap between them in points. N = 100,801, except the untouched validation season at 29,137. The bold row uses the previous gameweek's archival snapshot, which was taken before the current match was played. The FPL-demand-free benchmark contains no ownership, transfer or price variable. Standard errors are clustered on player and on gameweek. Appendix J reports nine further specifications.

Table 4

Calibration of the Demand Response to Predictive Content, by Signal Innovation

Innovation	Predictive content γ_j	Demand response β_j	Calibration residual $\beta_j - \hat{\lambda}\gamma_j$
Goals	0.0295	0.1118	+0.1051
Bonus points	-0.0061	0.0614	+0.0628
Bonus point index	-0.0117	0.0448	+0.0474
Assists	0.0129	0.0428	+0.0399
Clean sheets	0.0061	0.0337	+0.0323
Saves	0.0208	0.0020	-0.0027
Playing time	0.1291	0.0242	-0.0050
Yellow cards	-0.0039	-0.0077	-0.0069
Chance quality	0.0584	-0.0002	-0.0134
Chance creation	0.0635	-0.0080	-0.0224

Note. Predictive content is γ_j from Equation (3) and demand response is β_j from Equation (2), estimated jointly on the same 63,156 player-gameweeks with a constant, price rank, and existing ownership rank. Innovations and both outcomes are standardized, so each coefficient is the standard-deviation change in its outcome associated with a one-standard-deviation signal innovation. The residual is $\beta_j - \hat{\lambda}\gamma_j$, where $\hat{\lambda} = 0.226$; positive values indicate a larger response than implied by the fitted proportional benchmark, and rows are ordered by it. The omnibus proportionality test is $W = 111.0$, $p < .001$. Appendix Table S19 reports uncertainty and multiplicity-adjusted comparisons; Appendix Table S17 reports first-stage performance. Chance quality and chance creation are the platform's expected goals and expected assists, the two signals that award no points.

Table 5

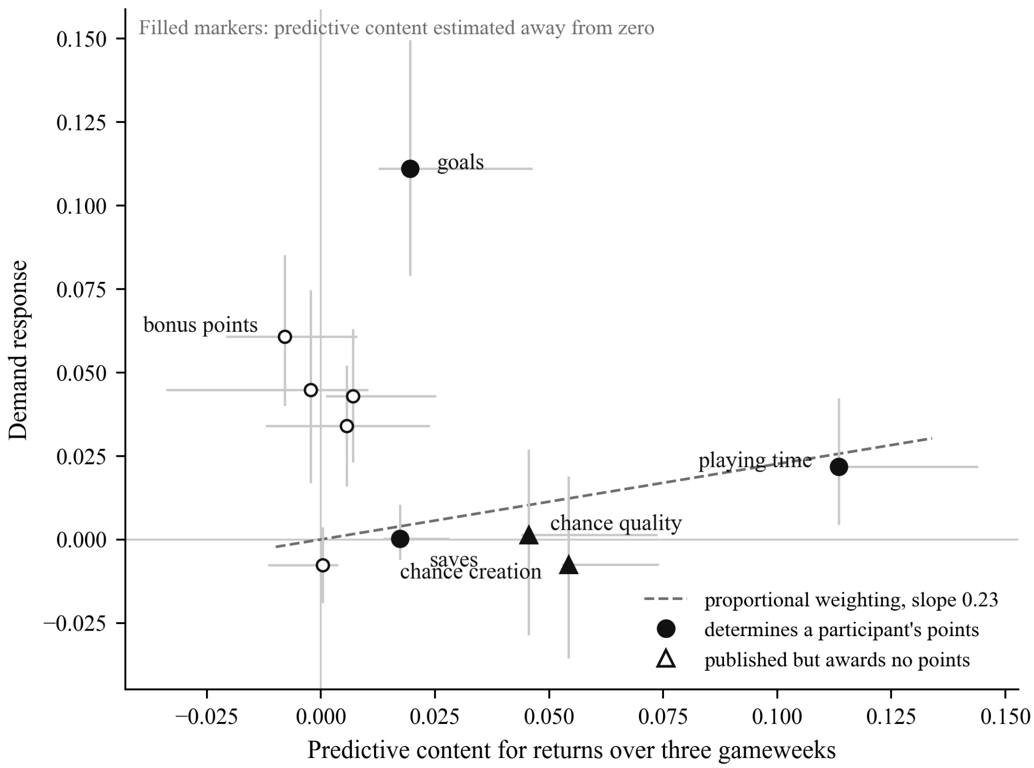
Ownership and Subsequent Chance Quality After Scoring, Matched Comparison

Test	Ownership change (pp)	t	Future chance quality, 3 GW (SD)
Main estimate	1.297	8.5	-0.003 (t = -0.09)
Player fixed effects added	1.256	7.3	0.005 (t = 0.15)
Goals from open play only	1.336	5.5	-0.003 (t = -0.08)
2024-25 evaluation season only	1.452	6.1	0.002 (t = 0.04)

Note. Ownership is the contrast in the subsequent change in ownership between players who scored and matched players who did not, in percentage points of registered entries. Chance quality covers gameweeks $t + 1$ through $t + 3$, in standard deviations. Groups are formed on match-level non-penalty expected goals, which also enters every regression linearly and quadratically. Group and gameweek effects are absorbed, and standard errors are clustered on player and on gameweek. Appendix Tables S9 to S15 report seven further constructions.

Figure 1

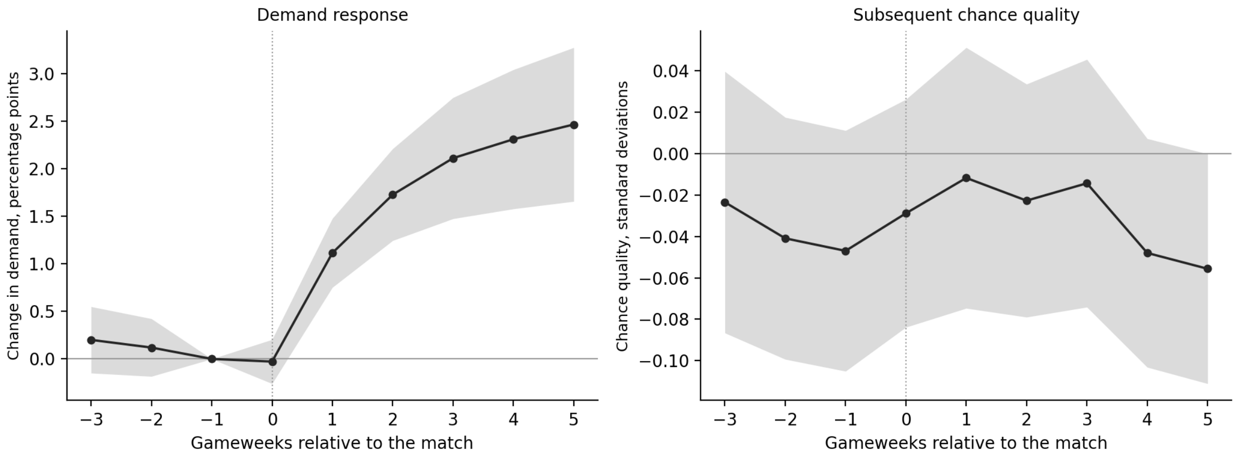
Demand Response Against Predictive Content, by Signal Innovation



Note. Horizontal axis, the predictive content γ_j of Equation (3); vertical axis, the demand response β_j of Equation (2), both on standardized innovations and standardized outcomes, $N = 63{,}156$. Marker shape distinguishes the eight point-relevant signals from the two the platform publishes but does not score; both are displayed to participants, so the distinction is scoring, not visibility. Filled markers indicate predictive-content estimates whose 95 percent confidence intervals exclude zero. The dashed ray is the benchmark of Equation (4) at $\lambda = 0.226$. Goals sit far above it; playing time, chance quality and chance creation far below and to the right.

Figure 2

Matched Ownership and Opportunity Dynamics Around Scoring



Note. Left panel, the contrast in ownership between players who scored and matched players who did not, in percentage points of registered entries, measured as a change from the gameweek before the match and therefore zero there by construction. Right panel, the contrast in subsequent chance quality, in standard deviations of its own distribution. Shaded bands are 95 percent intervals from two-way clustering on player and on gameweek. Gameweek zero is the match in which the outcome occurred.

Online Appendix

Appendices A to F state the assumptions behind the paper's tests; Appendices G to P document the data, the benchmark, matching, inference, robustness, the signal analysis, the auxiliary designs, the individual panel, the validation season, and reproduction. They are supplied as a separate file.

Data Availability and Code Replication

The study combines several data sources, which differ in their redistribution conditions.

Platform data. Player-gameweek points, minutes, prices, ownership percentages, platform expected goals and expected assists, fixtures, and deadlines come from the official Fantasy Premier League API (https://fantasy.premierleague.com/api/), as do the timestamped transfers of Section 3.3 and the 2025-26 validation panel. The archive provides the acquisition and construction code, source documentation, and checksums; platform-derived data are redistributed only to the extent permitted by the applicable platform terms. Entry identifiers are replaced by salted hashes before analysis and the salt is not distributed; release of row-level participant trajectories is governed by the applicable platform and privacy requirements.

Third-party data. Match-level non-penalty expected goals, expected assists, shots, and key passes come from Understat (https://understat.com/) and support the matched comparison of Sections 4.4 and 5.3 and the chance measures in Table 2. Match odds come from a commercial closing-odds feed and enter the alternative benchmarks of Section 5.1 and one auxiliary robustness specification in Section 5.3; no headline result in Sections 5.2 to 5.4 depends on them, and the FPL-demand-free benchmark of Table 3 is the specification for which they are load bearing. Appendix M.2 additionally uses Understat shot events from a public Kaggle archive released under the MIT license (Tipton, 2024), retaining fixtures between two Premier League clubs from 2021-22 to 2023-24; none of the paper's headline estimates depends on them. None of these sources is redistributed. The archive provides acquisition scripts, the identity maps used for the join, and checksums for each, so a replicator can reconstruct and verify the inputs.

The replication package is deposited under a persistent identifier, cited on acceptance and available to referees during review through an anonymized link. It includes an exact dependency lockfile, input checksums, and verification harnesses that recompute the reported results and flag inconsistencies between the analysis outputs and the manuscript.